

Problem Set 2

Applied Stats/Quant Methods 1/Nour Kebbi/ Answer Sheet

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.¹ As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, “We can solve this the easy way” to draw a bribe). The table below shows the resulting data.

¹Fried, Lagunes, and Venkataramani (2010). “Corruption and Inequality at the Crossroad: A Multi-method Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

- (a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```

1 #question 1 part (a)
2 #calculating test statistic that is distributed according to the x2
  distribution yb hand
3 #defining vectors
4 #calculating fobserved & fexpected
5 upper_class <- c("Not stopped" = 14, "Bribe requested" = 6, "Stopped" =
  7)
6 lower_class <- c("Not stopped" = 7, "Bribe requested" = 7, "Stopped" = 1)
7 upper_class_row_total <- sum(upper_class)
8 lower_class_row_total <- sum(lower_class)
9 not_stopped_column <- upper_class["Not stopped"] + lower_class["Not
  stopped"]
10 bribe_requested_column <- upper_class["Bribe requested"] + lower_class["
  Bribe requested"]
11 stopped_column <- upper_class["Stopped"] + lower_class["Stopped"]
12 grand_total <- sum(upper_class_row_total+lower_class_row_total)
13 #calculating the expected values
14 fe_not_stopped_uc <- (upper_class_row_total*not_stopped_column)/grand_
  total
15 fe_not_stopped_lc <- (lower_class_row_total*not_stopped_column)/grand_
  total
16 fe_bribe_requested_uc <- (upper_class_row_total*bribe_requested_column)/
  grand_total
17 fe_bribe_requested_lc <- (lower_class_row_total*bribe_requested_column)/
  grand_total
18 fe_stopped_uc <- (upper_class_row_total*stopped_column)/grand_total
19 fe_stopped_lc <- (lower_class_row_total*stopped_column)/grand_total
20 #putting the expected values in a vectors for each class
21 expected_uc <- c(fe_not_stopped_uc, fe_bribe_requested_uc, fe_stopped_uc)
22 expected_lc <- c(fe_not_stopped_lc, fe_bribe_requested_lc, fe_stopped_lc)
23 #putting all observed and expected values in vectors each
24 observed_values <- c(upper_class, lower_class)
25 expected_values <- c(expected_uc, expected_lc)
26 #calculating test statistic
27 x2_test_statistic <- sum(((observed_values-expected_values)^2)/
  expected_values))
28 x2_test_statistic

```

The test statistic is 3.791168.

- (b) Now calculate the p-value from the test statistic you just created (in R).² What do you conclude if $\alpha = 0.1$?

```
1 #calculating the expected_value
2 #calculating the degrees of freedom, rows-1 * columns-1
3 p_value <- pchisq(x2_test_statistic, df = 2, lower.tail=FALSE)
4 #p_value
```

Since the p-value is 0.1502306 and $\alpha = 0.1$ then p-value is greater than alpha and hence, we fail to reject the Null Hypothesis that the variables are statistically independent.

- (c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.322	-1.642	1.523
Lower class	-0.322	1.642	-1.523

```
1 #calculating the standarized residuals for each cell
2 row_proportion_uc <- upper_class_row_total/grand_total
3 row_proportion_lc <- lower_class_row_total/grand_total
4 not_stopped_column_proportion <- not_stopped_column/grand_total
5 bribe_requested_column_proportion <- bribe_requested_column/grand_total
6 stopped_column_proportion <- stopped_column/grand_total
7 #putting the proportions into vectors
8 row_proportions_vector <- c(rep(row_proportion_uc, 3), rep(row_proportion
  _lc, 3))
9 col_proportions_vector <- rep(c(not_stopped_column_proportion, bribe_
  requested_column_proportion, stopped_column_proportion), 2)
10 variance <- expected_values * (1-row_proportions_vector) * (1-col_
  proportions_vector)
11 se = sqrt(variance)
12 standardized_residuals <- (observed_values-expected_values)/se
13 z_matrix <- matrix(
14   standarized_residuals,
15   nrow = 2,
16   byrow = TRUE,
17   dimnames = list(
18     c("Upper Class", "Lower Class"),
19     c("Not Stopped", "Bribe Requested", "Stopped"))
20 )
21 print(round(z_matrix, 4))
22 #data <- rbind(upper_class, lower_class)
```

²Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

```
23 #chisq.test(data)
24 #test <- chisq.test(data)
25 #test$stdres
```

(d) How might the standardized residuals help you interpret the results?

We can see from the standardized residuals that upper class drivers are more likely to be stopped given the residual is 1.523 than being asked for a bribe (-1.642) or than not being stopped (0.322). We can also see that lower class drivers are more likely to be asked for a bribe given the residual is 1.642 than being stopped (-1.523) or than not being stopped (-0.322).

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.³ Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: <https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv>

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

³Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
GP	An identifier for the Gram Panchayat (GP)
village	identifier for each village
reserved	binary variable indicating whether the GP was reserved for women leaders or not
female	binary variable indicating whether the GP had a female leader or not
irrigation	variable measuring the number of new or repaired irrigation facilities in the village since the reserve policy started
water	variable measuring the number of new or repaired drinking-water facilities in the village since the reserve policy started

(a) State a null and alternative hypothesis (two-tailed).

Ho null hypothesis: having a reservation policy (female) head has no effect on the number of new/repared drinking water facilities in villages. The alternative hypothesis: having a reservation policy (no female) head has a positive or negative effect on the number of new/repared drinking water facilities in villages.

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```

1 #bivariate regression
2
3 my_table <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/
4   master/PREDICTION/women.csv")
5 head(my_table)
6 #calculating the means for each
7 mean_water_Y <- mean(my_table$water)
8 mean_reservation_X <- mean(my_table$reserved)
9 #calcuolating sums
10 sum_water_Y <- sum(my_table$water)
11 sum_reservation_X <- sum(my_table$reserved)
12 #claculating numerators for each
13 num_w_Y <- my_table$water - mean_water_Y
14 num_r_X <- my_table$reserved - mean_reservation_X
15 num <- sum((num_w_Y)*(num_r_X))
16 #calculating denominator
17 denom <- sum((num_r_X)^2)
18 #calculating beta
19 beta <- num/denom

```

```

20 alpha = mean_water_Y - (beta*mean_reservation_X)
21 alpha
22 model <- lm(water ~ reserved, data = my_table)
23 summary(model)

```

Beta is 9.252423 and Alpha is 14.73832

(c) Interpret the coefficient estimate for reservation policy.

Beta is 9.252423 and alpha is 14.73832 and beta now is 0 so villages with female

heads (reservation policy) have more drinking water, around +9.252423 than villages without female heads. On average, a female head reservation policy leads to a 9.252423 increase in drinking water. The Alpha is 14.73832, so villages without female heads have around 14.73832 new/repared drinking water. When there is no reservation policy (no female heads), the predicted value of new / repared drinking water is 14.73832. Note that it may not be a meaningful quantity.